This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Detection of Fake Accounts on Social Media Using Multimodal Data With Deep Learning

Bharti Goyal, Nasib Singh Gill, Preeti Gulia, Om Prakash, Ishaani Priyadarshini, Rohit Sharma, *Senior Member, IEEE*, Ahmed J. Obaid, and Kusum Yadav

***Abstract*— In recent years, the proliferation of fake accounts on social media has become a significant concern for individuals, organizations, and society. Fake accounts play an important role in spreading fake news, rumors, spam, unethical harassment, and other mischievous motives on social media platforms. Detecting such accounts manually is time-consuming and challenging, especially with the increasing sophistication of the methods used to create them. Therefore, there is a need for automated approaches to detect these fake accounts. This article aims to reveal fake accounts and their role on social media platforms. This article has explored various methodologies to detect fake accounts from social platforms, which will help prevent cybercrime. We have also proposed a generalized deep learning (DL) model to detect such accounts on social media using multimodal data. The proposed architecture uses a combination of textual, visual, and network-based features to capture the various characteristics of fake accounts. Specifically, we use a deep neural network that combines convolutional neural networks (CNNs) for visual data, long short-term memory (LSTM) networks for textual data, and convolutional graph networks (GCNs) for network-based data. We evaluated our model on a publicly available dataset of Twitter accounts and achieved state-of-the-art performance in detecting fake accounts, with an $F1$ score of 0.96. We also conducted experiments to show the effectiveness of each feature and the combination of the three features.**

***Index Terms*— Artificial neural networks (ANNs), convolutional neural networks (CNNs), fake account, machine learning (ML), social media.**

Manuscript received 25 February 2023; revised 8 April 2023, 2 May 2023, 16 June 2023, and 5 July 2023; accepted 13 July 2023. *(Corresponding author: Rohit Sharma.)*

Bharti Goyal, Nasib Singh Gill, and Preeti Gulia are with the Department of Computer Science and Applications, Maharshi Dayanand University, Rohtak 124001, India (e-mail: bharti.rs.dcsa@mdurohtak.ac.in; nasib.gill@mdurohtak.ac.in; preeti@mdurohtak.ac.in).

Om Prakash is with the School of Computer and System Sciences, Jawaharlal Nehru University, New Delhi 110067, India (e-mail: omprak16_scs@jnu.ac.in).

Ishaani Priyadarshini is with the School of Information, University of California, Berkeley, Berkeley, CA 94720 USA (e-mail: ishaani@berkeley.edu).

Rohit Sharma is with the Department of Electronics and Communication Engineering, SRM Institute of Science and Technology, Delhi NCR Campus, Ghaziabad 201204, India, and also with the Scientific Research Center of the Islamic University, The Islamic University, Najaf 192122, Iraq (e-mail: rohitapece@gmail.com).

Ahmed J. Obaid is with the Department of Computer Science, Faculty of Computer Science and Mathematics, University of Kufa, Kufa 540011, Iraq, and also with the Department of Computer Technical Engineering, Technical Engineering College, Al-Ayen University, Thi-Qar 46477, Iraq (e-mail: ahmedj.aljanaby@uokufa.edu.iq).

Kusum Yadav is with the Department of Computer Science and Applications, Maharshi Dayanand University, Rohtak 124001, India, and also with the Department of Information and Computer Science, College of Computer Science and Engineering, University of Ha'il, 2240 Ha'il, Saudi Arabia (e-mail: y.kusum@uoh.edu.sa).

Digital Object Identifier 10.1109/TCSS.2023.3296837

I. Introduction

THE century we live in is known as the era of the digital world. In this era, with the rapidly expanding digital environment, the development of society, and the rise of technology, the online population is increasing daily. With the invention of the Internet, communication has become easy and fast without time and space bound. Online social media is one of the results of this. Social media has become integral to our lives, enabling us to connect and communicate with people worldwide [1]. Nowadays, individuals use various social media platforms to communicate with each other. So online social media is the fastest growing network in the world. With the advancement in technology, there is massive growth in the number of users using online social media platforms, which causes a security risk to data available on these platforms, the proliferation of fake accounts is one of them. Fake accounts can be used for various purposes, such as spreading misinformation [2], propaganda, and even for financial fraud. Fake accounts create many issues on online social media platforms, including online hacking, fake news about the government during COVID-19 pandemic, spam bots, misinformation spread through websites, fake pages of organizations, fake accounts of celebrities, etc. Detection of such accounts is essential to retain social media platforms' credibility.

In recent years, researchers have proposed various methods to detect fake accounts on social network platforms, including using features such as user metadata, network-based features, and content-based features. However, these methods often rely on a single feature type and do not consider the multimodal nature of social media data, including textual, visual, and network-based data.

This article aims to detect fake accounts. As we already discussed, such accounts make nuisances in society, such as cyber threats and crimes, and play a significant role in spreading fake news, rumor, spam, and unethical harassment on social media platforms. Detection of such profiles is essential to retain social media platforms' credibility. So, by detecting such accounts, this article contributed to putting a stop to cyber-crime leading to a healthy online social network society having fear-free life on such platforms. In this research, we investigated many methods for identifying phoney social media accounts, which will aid in the prevention of online crimes perpetrated using such accounts. In addition, we have put forth a deep learning model that uses multimodal data to identify fake social media accounts. The suggested design captures the many traits of phoney accounts using a mix of

2329-924X © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

textual, visual, and network-based information. We use a deep neural network composed explicitly of convolutional graph networks (GCNs) for network-based data, long short-term memory (LSTM) networks for textual data, and convolutional neural networks (CNNs) for visual data. With an $F1$ score of 0.96, our model, tested on a publicly accessible dataset of Twitter accounts, performed at the cutting edge of fake account detection. We also ran studies to demonstrate the usefulness of each feature alone and when combined. Our proposed approach has significant implications for social media platforms, law enforcement agencies, and researchers in social media analysis. Using multimodal data with deep learning, our approach can help detect fake accounts on social media with high accuracy, which can help mitigate the negative impact of these accounts on individuals and society. The rest of this article is organized as follows. Section II reviews related work, Section III describes the proposed method, Section IV presents the experimental setup and results, and Section V concludes the article and discusses future work.

The main objectives of this study are as follows.

1) This study provides a detailed overview of fake accounts by describing their role on online social media.
2) This article discusses the impact of fake accounts on social media by describing how they influence social media vulnerability.
3) This study outlined the best machine learning (ML) approach to detect fake accounts and proposed a generalized deep learning approach to detect fake accounts on Twitter, which can be further used on other social media platforms.

II. Fake Account Detection Techniques on Social Media

Detecting fake accounts on social media has become increasingly important due to the negative impact these accounts can have on individuals and society. In recent years, researchers have proposed various methods for detecting fake accounts on social media, including using features such as user metadata, network-based features, and content-based features. However, these methods often rely on a single feature type and do not consider the multimodal nature of social media data, including textual, visual, and network-based data. To address these limitations, several studies have proposed methods for detecting fake accounts on social media using multimodal data with deep learning. These studies have shown that combining different data types can improve the accuracy of fake account detection. Bovet and Makse [3] found a way to find fake Twitter accounts using network-based and content-based features. The study used a deep neural network that combined a GCN for network-based features and a CNN for content-based features. The results showed that the proposed method outperformed previous state-of-the-art methods.

Wang et al. [4] proposed a method for detecting fake accounts on social media using a combination of textual and visual features. The study used a deep neural network that combined an LSTM network for textual features and a CNN for visual features. The results showed that the proposed method found fake accounts correctly in a publicly available dataset. Arora et al. [5] devised a way to find fake social media accounts using a mix of textual, visual, and network-based features. The study used a deep neural network that combined an LSTM network for textual features, a CNN for visual features, and a GCN for network-based features. The results showed that the proposed method was the best way to find fake accounts in a publicly available dataset.

Overall, these studies show that it is possible to find fake social media accounts using multimodal data and deep learning. Combining different data types can capture different characteristics of fake accounts, such as the language used in the account's posts, the visual content shared by the account, and the network structure of the account's followers and friends. These methods significantly affect social media platforms, law enforcement, and social media analysis researchers.

A. Detection Techniques

There are various techniques for detecting fake accounts on social media, including metadata analysis: This method involves looking at the information about an account, such as when it was made, where it is, and how it is used. Fake accounts are often made in groups, and their metadata are often inconsistent, which can be used to spot them.

1) Content-Based Analysis: This method involves looking at the text, images, and videos on account posts. Fake accounts often use generic or plagiarized content, have a high posting frequency, and tend to use spammy or clickbait headlines. Network-based analysis: This method involves looking at an account's network of friends and followers. Fake accounts often have few followers, and the followers are often fake or bot accounts.

2) Machine Learning: This method teaches machine learning models to spot fake accounts based on metadata, content, and network-based features. Models that use machine learning can learn to recognize patterns that are hard for humans to see and can be very good at finding fake accounts.

3) Hybrid Approaches: Some studies have suggested hybrid approaches that combine multiple techniques, such as metadata analysis and content-based analysis or machine learning, to make it easier to spot fake accounts.

The best ways to find fake social media accounts combine metadata, content, and network-based features with machine learning to analyze large datasets. These techniques can help stop the spread of false information and harmful content on social media, essential for social media platforms, law enforcement agencies, and researchers in social media analysis. Previous work to detect fake profiles using various techniques includes the following.

Zhou et al. [6] proposed an algorithm to detect fake profiles on a recommender system. The authors introduced the time stamps target item analysis (TS-TIA) algorithm. This method uses two statistical metrics and a time series of attack ratings to find odd profiles based on how attacks are rated. Zeng et al. [7] proposed a model to detect users on Twitter and Facebook. A system multiple user information user identification framework (MUIUI) for determining users based on their user information was used. First, they made multiprocess

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

crawlers to get user data from Facebook and Twitter. Second, they used entity linking and entitled entity recognition to get locations from profiles and user-generated content (UGC), which they then put together. In addition, URLs from user profiles and UGC were retrieved. Locations were used in conjunction with the relationships, and several algorithms were constructed. A user identification approach based on fusion classifiers and machine learning was presented. The findings revealed that the MUIUI $F1$ score reached 86.46%. It is concluded that MUIUI is capable of lessening the influence of user data that are either absent or incomplete.

Mohammadrezaei et al. [8] proposed an approach to detect fake accounts on social media platforms based on the similarities of the user's friends. In this methodology, the friend similarity criteria were initially computed using the adjacency matrix of the network graph. Then additional features were recovered using the principle component analysis (PCA) approach. In the next step, the synthetic minority oversampling technique (SMOTE) was used to normalize the data before it was sent to the classifier. The fictitious users were predicted by computing similarity and using classifier methods. Yang et al. [9] proposed an approach to detect social spam bots on social networks. Profile and time information were used to find social bots. This method eliminated the need for tedious feature engineering and could build features automatically, so it could keep up with how quickly social bots were changing. The method focused on the content of postings as temporal data. The bidirectional encoder representation from the transformers (BERT) language model was used to extract the semantic information of the text, and the BiGRU algorithm was used to extract the latent temporal patterns between the postings to portray the social bot in OSNs.

Siino et al. [10] did a comparative review of state-of-the-art (SotA) models, explicitly focusing on Transformers, to solve the challenge of detecting fake news spreaders. Corpus linguistics techniques were used. According to the results, big, pretrained deep models, such as Transformers, are sometimes the best option when solving a text classification problem. Bertini et al. [11] proposed an approach to detect fake profiles on social media platforms based on image watermarking. It first studied how social networking sites handle photographs posted by users to discover a potential application of image watermarking techniques. Secondly, 13 social networks are used to evaluate the resiliency of several different picture watermarking systems. Finally, it is determined whether digital forensic or image forgery detection activity using the photo response nonuniformity (PRNU) technology can be used effectively as a watermarking strategy for authorship identification and verification of images on social networks. The suggested method resolved issues such as profile linkage and the identification of phoney profiles. In addition, the approach made it possible to define multifactor online authentication methods founded on secure digital characteristics.

Egele et al. [12] illustrated how similar approaches might be used to discover breaches of personal accounts related to prominent organizations. The authors proposed a COMPA approach to detect compromised accounts and stop three actual cyberattacks launched against well-known businesses and news organizations. COMPA used statistical models to define netizens' behavior and anomaly detection techniques to spot the abrupt behavioral changes in these users. COMPA used statistical models to define netizens' behavior and anomaly detection techniques to spot the abrupt behavioral changes in these users. Wu et al. [13] proposed an approach to protect users' personal information when using a recommender systems service. The authors used a client-based system architecture that does not involve changes to the already-existing recommendation algorithms and does not impact the precision of the outcomes of any recommendations made. Finally, the proposed approach can create a group of high-quality fake recommendation profiles, which have similar characteristics to the actual user preference profile to hide the actual profile, as well as the potential to lower the risk of disclosing the user's sensitive data.

Qu et al. [14] proposed a Sybil detection model called MUSH to identify long-term user behavior characteristics on user-reviewed social networks (URSNs). Four entropy-based preference models were built for catering, pricing, rating, and word-of-mouth preferences. Then multistimuli Hawkes method was developed, a combination of internal and external incentives, and it can identify unusual occurrences of reviews posted. Finally, a Sybil detection model based on the binary classification technique was developed. MUSH proved an excellent choice in the adversarial network by improving the quality of security services to social media platforms. Lin et al. [15] proposed an attack model called learning to generate fake user profiles (Leg-UP) to understand shilling attack approaches. Leg-UP builds phoney user profiles by gaining knowledge about user activity patterns from real users. The generator in Leg-UP creates discrete ratings to replicate the behavior of real users. To improve attack transferability, the generator parameters were optimized by getting the best attack performance on a stand-in Reed Solomon (RS) model. In addition, Leg-UP used a discriminator to direct the generator in creating fake user profiles that could not be detected, increasing attack invisibility. Tests on victim RS models demonstrated that Leg-UP is superior to most cutting-edge shilling attack techniques.

Moghaddam and Abbaspour [16] proposed a feature named friendship preference to discriminate the bot from real user accounts on online social media platforms. The profile properties of the followers were mined for information on friendship preference aspects. If the number of followers grows, the feature extraction methodology should be scalable. Two different classifiers were built to assess the suggested category of characteristics in respective levels of utility and efficiency. The classifiers were evaluated on several different real datasets. According to the experiments' findings, the classifiers' performance was superior to Botometer. The scalability of a system was evaluated by observing how well classifiers identify anomalies when just the characteristics of a small subset of the accounts located close to users with huge geonets. In addition, the generalization was validated by performing cross-validation on different datasets. Bharti and Gulia [17] discussed various techniques for fake account detection on online social networks. In the same way, Ersahin et al. [18]

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

proposed classification method to detect fake accounts on Twitter. Bharti and Pandey [19] used logistic regression with particle swarm optimization for Twitter fake account detection. Borkar et al. [20] using recurrent neural network (RNN) detected Twitter fake accounts. Prakash and Kumar [21] detected Twitter fake accounts using supervised machine learning (SML).

Hasan et al. [22] proposed a model for keeping an eye on rising costs and potential instances of bribery in the Industry 4.0 age. This approach incorporated distributed ledger technology (blockchain), cloud servers, relational database management systems (RDBMS), and remote database access (RDA) protocols. This model cautioned management to safeguard small businesses against unrestrained data changes in situations where applications were connected to RDBMS that used the RDA protocol. Diro et al. [23] suggested a system to provide a lightweight solution while outsourcing calculations and storing for security parameters, which makes use of a fog server as an intermediary broker. The suggested system reduces the number of handshakes performed during each resumed session. Ahmed et al. [24] developed a model to limit the spread of COVID-19 pandemic infection, a social distance surveillance structure based on the principles of deep learning architecture. The proposed model achieves 96% accuracy. Ahmed et al. [25] used an overhead perspective to demonstrate a social distance monitoring framework. This study used transfer learning for a deep-learning-based detection paradigm. The recognition model provides information regarding the bounding area, including the centroid coordinates. A violation criterion is used to determine whether or not the distance value breaches the minimal social distance that has been established.

Zhang et al. [26] propose a technique of complex analysis to disclose the variables that influence the user engagement of the metaverse from the point of view of social patterns and reward mechanisms. Priyadarshini et al. [27] developed a method called transfer learning, which incorporates two different machine learning models to identify offensive and hateful expressions on social media platforms effectively. The research was carried out on several different datasets, and the findings appear consistent across all the datasets. The performance of the machine learning models is evaluated in terms of recall, precision, $F1$ scores, and support. The machine learning models were responsible for corresponding tweets into the neural, offensive, and hatred categories. Both accuracy and confusion matrices have been used to validate the model's performance. The experimental analysis results indicate that the suggested model achieves a higher level of precision than any of the other baseline models. Vo et al. [28] analyzed Twitter data collected from Twitter accounts for crime discovery. Analysis of users' behavior and personality is done through their tweets using sentiment analysis, which has been used to trace criminal activity by monitoring tweets. To identify parts of speech in online conversational writing, the Twitter part-of-speech tagger, a Markov model of first-order entropy, has been used. When there are a large number of tweets that are not identified, brown clustering is used. The comparisons are checked against actual crime statistics from a reliable data source according to various geographical locations. The analyses show that the acquired results are consistent with the statistics regarding the actual crime rate.

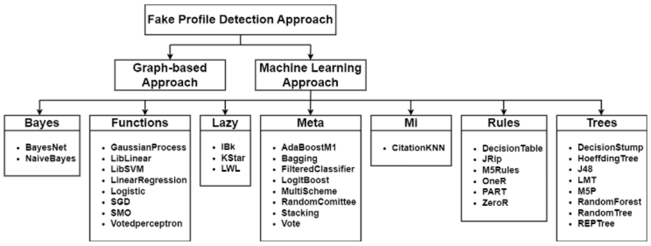

Fig. 1. Fake profile detection approaches.

Kwek et al. [29] researched fake news during COVID-19. This research provides a comprehensive knowledge of how social and cultural factors (collectivism) contribute to constructing judgments and adaptations among Internet users regarding fake news. Tiwari et al. [30] proposed a model named multiagent federated reinforcement learning policy, also known as MFRLP, for cyborg detection. It was indicated in mobile and fog agents and was based on the socket remote procedure call (RPC) paradigm. The objective was to create a training and testing system that was consumer-focused, cyborg-efficient, and capable of executing the complete application process with the fewest possible delays in the IoMT system. For use in cyborg applications, the research proposed an RPC based on reinforcement learning and federated learning that adopted the dynamic changes in the surrounding environment. Ghazvini et al. [31] proposed a model to predict commercial serial crime. In light of this, the authors proposed the enhanced NARX nonlinear autoregressive with exogenous input (eNARX) model. This model was based on fusing two functions in the same hidden layer: the hyperbolic tangent (Tansig) and the radial basis function (RBF). They used five different time-series datasets to test the suggested method and compared it with the already established NARX. These datasets included the Dow Jones Index, the Monthly River flow in cubic meters per second, the Daily temperature, and two UKM-PDRM datasets called "Suspect & Capture" and "Crime Plotting." The investigation of the obtained information revealed that the suggested eNARX generates more accurate results than those generated by previous NARX methodologies.

Different approaches used for fake profile detection are shown in Fig. 1.

Fig. 1 represents that fake profile detection approaches include graph-based and DL approaches, where the machine learning approach is further divided into categories including Bayes, Functions, Lazy, Meta, Mi, Rules, and Trees.

B. Impact of Fake Accounts on Social Media

As fake accounts on social media can have a significant impact on individuals and society as a whole, some of the major impacts include.

1) Spread of Misinformation: Fake accounts can be used to spread false information and propaganda, which may have severe consequences for individuals and society. This can lead

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

to confusion, mistrust, and even harm, such as the spread of false information about vaccines [32] or elections [33].

2) Cyberbullying: Fake accounts can be used to bully, harass, or intimidate individuals online. It harms mental health and can lead to real-world harm, such as online harassment [34], leading to physical violence.

3) Fraud: Fake accounts can be used to perpetrate fraud or scams, such as phishing attacks [35] or fake charity fundraisers [36]. This can result in financial losses for individuals and organizations.

4) Manipulation of Public Opinion: Fake accounts can be used to manipulate public opinion by artificially inflating the popularity of specific individuals, ideas, or products [37]. This can significantly impact political and economic [38] outcomes.

5) Privacy Violations: Fake accounts can be used to collect personal information from individuals or to impersonate [39] them online, which can result in privacy violations and identity theft. Table I illustrates the different roles of fake accounts on different social media platforms.

Overall, the impact of fake accounts on social media can be significant and far-reaching. It is important for social media platforms, law enforcement agencies, and individuals to be aware of the risks associated with fake accounts and to take steps to prevent their proliferation and misuse.

III. Dataset

Data collection and preprocessing are essential parts of any machine learning project, such as using multimodal data and deep learning to find fake social media accounts. The quality and amount of the data collected and the steps taken before building the model have a significant effect.

A. Data Collection

Due to the size of the data and the fact that it comes from many different formats and sources, collecting data from social media can be challenging. However, social media platforms often have application programming interfaces (APIs) that can be used to collect data in a structured and organized way. Here are some common ways to collect data from social media.

1) APIs: Social media sites such as Facebook, Twitter, and Instagram allow developers to access their data programmatically.

2) Web Scraping: Web scraping involves extracting data from websites by analyzing the HTML and Cascading Style Sheet (CSS) code. However, web scraping can take much time and put much stress on the servers that host the website.

3) Crawling: Crawling involves collecting data by following links from one page to another. This method can help collect data from social media networks that do not provide APIs and collect data from many sources.

The dataset is prepared in our work using the Twitter API [51] for collecting social media data from Twitter. The dataset contains 12 355 accounts, out of which 8267 are genuine accounts and 4088 are fake accounts. The dataset is obtained, features are analyzed, and relevant features are extracted. This means cleaning the data, filtering the unavailable information, and putting the data in a format that can be performed in our machine learning approach. Fig. 2 shows the features of the Twitter user profile, which have various features from the user metadata and profile information.

TABLE I
Fake Account Roles in Social Media

Key Reference	Fake Account Role Description
Y. Boshmaf et al. [40]	Fake befriend accounts connect with real accounts by sending friend requests and steal useful information from the victim.
M. Vyawahare et al. [41]	Users create fake accounts using gender as the main factor to attract the audience. Gender and profanity factors proved good in fake profile detection.
S. Adikari et al. [42]	Attackers target LinkedIn, a professional social media platform, to steal user information from genuine accounts.
J. B. Munga et al. [43]	Fake users on Facebook play a significant role to steal user information and spread fake news causing threats to society.
M. Orabi et al. [44]	Attackers gain access to social media platforms by creating Social Media Bots, which spread fake news, rumours, etc., on social media platforms, especially Twitter, to gain popularity.
K. S. Adewole et al. [45]	Spam accounts, compromised accounts, etc. types of accounts give rise to phishing, spamming, malware, etc., challenges to society.
K. S. Adewole et al. [46]	Fake profiles on Twitter work as spammers to spread false information and do some malicious activities.
D. Ramalingam et al. [47]	When engaging in nefarious behaviour on social media sites, fake profiles are essential for many reasons, including but not limited to any particular area.
O. Prakash et al. [48]	Fake profiles give rise to cybercrime by spreading fake news on social media platforms.
Q. Ye et al. [49]	Fake Accounts play an essential role in financial fraud in e-commerce.
Y.-C. Wei et al. [50]	Fake profiles make money from online games giving growth to the online game market.

id	name	profile_link_color	default_profile_image
followers_count	screen_name	profile_text_color	profile_use_background_image
created_at	utc_offset	profile_sidebar_border_color	profile_background_tile
geo_enabled	description	profile_sidebar_fill_color	profile_background_image_url_https
friends_count	updated	profile_background_color	profile_banner_url
protected	verified	status	profile_image_url_https
			profile_background_image_url
			url

Fig. 2. Twitter user profile features.

B. Data Preprocessing and Feature Selection

Data preprocessing is an essential step in any machine learning project and involves cleaning and transforming the

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Fig. 3. Unified deep learning architecture of the proposed model.

data into a format suitable for analysis. Here are some common data preprocessing. Now the data collected were ready to be used. The next step required was the feature selection to experiment. Now every attribute works as nominal data to the applied techniques. Among all the features, "fake" is selected for binary classification to simulate the experiment.

IV. Proposed Methodology

A generalized ML architecture is a systematic way of constructing and developing an ML model that can correctly and quickly identify fake or manipulated social media accounts. The proposed architecture consists of several submodels, including data collection, preprocessing, feature engineering, data fusion, generalization and optimization, deep learning techniques for classification, and an output submodel, as shown in Fig. 3.

Fig. 3 tells us about the flowchart steps involved in preparing the final output from the data collection step. Here at first, data are collected, and then as data are not in the proper format to be used, they are preprocessed. The next step is data fission and optimization. Then various machine learning techniques are applied, generating the final output.

The architecture must be flexible enough to find accounts that have been hacked or are not real across different social platforms. The goal of this architecture is to make a model that is stronger and can use the best parts of multiple DL models. This architecture will make it easier to build accurate, scalable, and reliable models to find and track accounts that have been hacked.

A. Unified Architecture

In this work, we propose a dense deep learning architecture to tell the difference between real and fake accounts. The model learns patterns from multimodal data and performs classification tasks, which pollute online social environments. It comprises five sub-blocks or modules, each with a clear functional purpose. The first block is data collection from Twitter API called Tweepy. The second block is called preprocessing and feature engineering; in this phase, we filter out any raw and noisy data that are not required, separating them into different "unimodality." In the data extraction process from the Twitter database, the metadata about the social profile are also available: location, followers, retweets, number of posts, profile image, banner image, tweet text stream, etc. The overall feature processing, selecting, filtering, transforming the essential feature, and dividing them into different unimodal streams are called feature engineering so that it is ready for supervised learning. In the next block in the architecture, different features from the three streams are fused, also called knowledge fusion, to form a high-dimensional feature to classify the social profile as fake or genuine status.

B. Data Collection and Preprocessing

Fig. 4 is the extended version with a detailed explanation of the unified architecture.

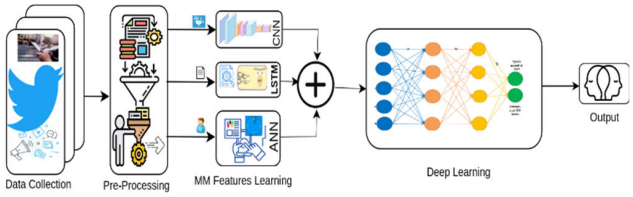

Fig. 4. Proposed unified-generalized machine learning model.

Fig. 4 is the extended version of Fig. 3, in which data are collected and then preprocessed to give it a structured form. In the third step, ANN, LSTM, and CNN are applied, and then using deep learning, the desired output is achieved.

To track and differentiate between manipulated and genuine accounts, we need to collect and analyze the profile's details; first, we need to gather all the related information about that particular account. This information can provide insights into various aspects of social behavior, including online communication, community engagement, sentiment analysis, and content sharing. Twitter API Tweepy is used for collecting social media data, including web scraping, APIs, and social media monitoring tools. It is essential to note that collecting social media data also raises ethical and privacy concerns. Preprocessing and feature extraction of the data are the next steps after collection. Here, we use various operations to remove functional attributes from each data modality. For example, we clean up textual data before passing it to LSTM, an RNN with better memory performance than conventional recurrent neural networks; the irrelevant information is deleted. Contrarily, following picture preprocessing, such as shrinking and normalizing the pixels, visual contents are turned into visual embedding vectors using CNNs. Similarly, we undertake a data cleaning procedure on statistical data by eliminating any missing or superfluous data and then converting the data using scaling. During the preprocessing step, raw data are changed into a format that can be used for further analysis or modeling. Preprocessing involves various techniques, such as cleaning, transforming, and reducing the data, feature scaling, feature selection, data fusion, and integration and reduction.

C. Multimodal Features' Learning

Multimodal feature learning is a technique used in machine learning to process and analyze data with input or features such as text, images, video, and audio obtained with Twitter's Tweetpy API. The proposed architecture extracts the multimodal data in three modes: numeric feature from the metadata data input to the neural network, and tweet description as textual data input to the language processing model, which tokenizes the text data and prepares it for the feature learning process. The visual data, like the profile picture and banner, are fed into the convolution model. It creates a visual cue that can tell whether a profile is accurate. Different ways to learn multimodal features exist, such as early, late, or multilevel fusion. In the proposed method, we used late fusion, which involves processing each modality separately and then combining the results of each modality to make the final decision.

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

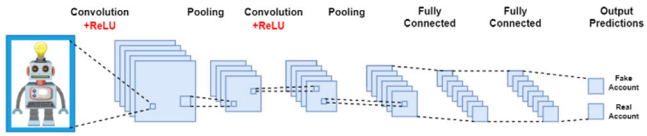

Fig. 5. Block diagram of CNN.

However, multimodal feature learning also presents challenges, such as handling missing or noisy data from one or more modalities, selecting appropriate fusion techniques, and dealing with the computational complexity of processing multiple modalities. Therefore, it is essential to carefully design the model and choose the appropriate techniques based on the specific application and dataset. In Fig. 4, the input for the machine learning approach is prepared in three modes. First, the proposed CNN module uses the visual data as the profile image and the banner image to learn the spatial feature from the image. In the second phase, LSTM is used to feed the tweet description that processes the ANN data, which are processed quantitative metadata such as the number of follows, likes, retweets, and mentions in the third mode.

1) Convolutional Neural Networks: CNNs are deep learning algorithms often used to recognize image patterns. This network is a regularized version of a fully connected network. Each layer is partially connected but sparsely connected with the next layer. A CNN learns visual patterns through different filters. CNN has many layers, but mainly two operations are used: 1) convolution and 2) pooling. *Pooling* is a method used in CNNs to reduce the spatial dimensionality of the input data. Usually, after a convolutional layer, a pooling layer acts separately on each feature map. The most popular pooling is max pooling, where the output of the pooling layer is the highest value found inside a particular area of the feature map. Other pooling forms include L2-norm pooling, which takes root mean square value within an area, and average pooling, which takes the average value inside a region. By reducing the spatial size of the feature maps by pooling, overfitting is prevented by lowering the number of parameters in the network. Making the feature maps more translation-invariant also aids in strengthening the network's resistance to slight changes in the input. However, pooling can also lead to information loss, especially if the pooling regions are too big. Stridden convolutions may accomplish a comparable decrease in spatial dimensionality while maintaining more of the original information. Therefore, in certain circumstances, they may be preferable to pooling. We have used the max pooling layer, where the output of the pooling layer is the highest value found inside a particular area of the feature map. This sequence of layers ensures learning high-dimensional feature space. The network contains fully connected layers in the last few layers. These fully connected layers are used for classifying or regaining learned representative variables. Fig. 5 shows a simple CNN architecture. CNN has performed exceptionally well in image recognition with these two layers. So, we have used only these two layers.

2) LSTM: LSTM is a unique neural network designed to prevent the long-term dependency problem in RNN. It is the evasion behavior of LSTM to remember information for an extended period. LSTMs can add or remove cell state information using the gates, which is the central idea behind the architecture, as shown in Fig. 6.

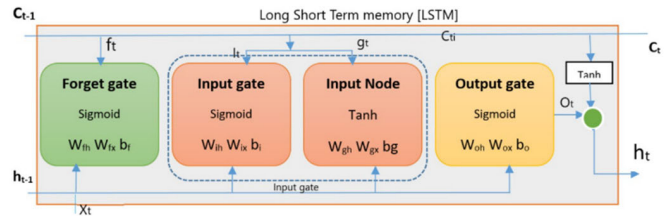

Fig. 6. Architecture of LSTM.

Fig. 6 shows the architecture of LSTM where forget gate, input gate, input node, and output gate are shown. By multiplying and adding, LSTMs make small changes to the information. With LSTMs, the information flows through a system called cell states. In this way, LSTMs can selectively remember or forget things that are important and not so important. There are three significant gates inside each cell: forget gate (ft), the input gate (it), and output gate (ot). Forget the gate and use the sigmoid function, as shown in (1), to determine what information escapes the cell. The sigmoid layer outputs a number between 0 and 1, describing how much each component should be let through. A value of 0 allows nothing, whereas 1 allows everything through. The next step is to store the new information in the cell, done by the input gate. It has two parts; first, it decides which value should be updated using sigmoid, as shown in (2). The second step uses the Tanh layer, creating a vector of new candidate values by (3). Furthermore, it combined both the steps to update the state as (4) described. Finally, the output gate uses a sigmoid layer, which decides what part of the cell will output and multiply with Tanh, as shown in (5), where σ represents the sigmoid function, $W*$ are the weight matrices for the corresponding dates, and $b*$ are biased. A dot represents the elementwise multiplication of two vectors

$$f_t = \sigma\left[\left(W_{fh} \cdot h_{t-1}\right) + \left(w_{fx} \cdot x_t\right) + b_f\right] \tag{1}$$

$$i_t = \sigma\left[\left(W_{ih} \cdot h_{t-1}\right) + \left(w_{ix} \cdot x_t\right) + b_i\right] \tag{2}$$

$$g_t = \mathrm{Tanh}\left[\left(W_{gh\cdot} \cdot h_{t-1}\right) + \left(w_{gx} \cdot x_t\right) + b_g\right] \tag{3}$$

$$C_t = ft \odot C_{t-1} + it \odot gt \tag{4}$$

$$o_{\mathrm{t}} = \mathrm{Tanh}\left[\left(W_{\mathrm{oh}} \cdot h_{\mathrm{t}-1}\right) + \left(w_{\mathrm{ox}} \cdot x_t\right) + b_o\right]$$

$$h_{\mathrm{t}} = o_t \odot \mathrm{Tanh}(C_t). \tag{5}$$

3) ANN: Artificial neural networks (ANNs) consist of layers of interconnected nodes, or neurons, which process and transmit information. The basic unit of an artificial neuron is the perceptron, which takes a set of inputs, multiplies them by weights, and adds a bias term. The result is then sent to an activation function, which decides whether or not the neuron is active. The output of a neuron is then sent to other neurons in the network through weighted connections. Several techniques, such as regularization and dropout, have been used.

4) Classification and Prediction: In the fourth step, the generalized multimodal feature vector is transferred to the deep learning submodule to discover hidden patterns in multimodal training data. The goal is to predict the class label

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Algorithm 1 To Compute Batch Normalization [53]

Input: Values of x over a mini-batch: $B = \{x_1, x_2, .., x_m\}$
Parameters to be learned: γ, β
Output: $\{y_i = BN_{\gamma,\beta}(x_i)\}$
$\mu_\beta \leftarrow 1/\ m\sum_{i=1}^{m} x_i$ // mini-batch Mean
$\sigma^2\beta \leftarrow 1\sum_{i=1}^{m} (x_i - \mu_\beta)^2$ // mini-batch variance
$\hat{x}i \leftarrow (x_i - \mu_\beta)/\sqrt{\sigma 2\beta + €}$ // normalize
$yi \leftarrow \gamma\ \hat{x}\ i + \beta = BN,\beta(x_i)$ // scale and shift

of given multimodal data based on features or attributes that describe the input. The input can be an image, audio clip, text document, or any other data type. The deep neural network is trained on a labeled dataset, in which each input is linked to a known class label. During training, the network weights and biases are changed so that the network learns to map the input features to the correct class label. Deep learning is a powerful technique for classification problems, with the ability to handle data outside of the distribution. The next and last step is output, where the classifier's output is converted into an easily understandable format.

D. Learning With Generalization Ability

In the regularization-based method, we used data augmentation, dropout [52], statistical regularization techniques, and other trivial techniques, while normalization-based approaches such as batch normalization (BN) are used for generalization. After extracting textual and visual information, we used BN. Next, we concatenate this vector of generalized characteristics before doing BN. We train the network weights using a mini-batch and the stochastic approach. A mini-batch is a subset of the sample input determined by the batch size. Total training relies on the initialization of the network and the random selection of mini-batch sizes. The BN is applied to a mini-batch to stabilize its mean and standard deviation. The procedure creates an atmosphere suitable for learning the parameters and may remove the need to pursue moving targets. The algorithm is computed by computing the batch means and variance and normalizing the activation described by the algorithm.

Here, μ is the mean, and σ is the standard deviation of the mini-batch. A different set learns two trainable parameters γ and β of mini-batch from input data, not independently. The BN maintains gradient flow in the network; hence, saturating nonlinearity is also used.

V. Results and Analysis

This research aims to create a deep learning model that uses data from multiple sources to find fake social media accounts, which are primarily responsible for spreading misinformation on social platforms. We used Twitter APIs to collect data from Twitter. We tested and visualized how well each feature worked on its own and how well it worked together. We used a local GPU-based system with a 4-GB graphics card and the latest Python package. For data analysis, the TensorFlow framework uses the Keras API and various Python packages, such as Sklearn, Pandas, Numpy, Matplotlib, and Tweetlib. We used several NNs and set hyperparameters to tune each NN's generalization capability. The high accuracy of our method has important implications for social media platforms.

TABLE II
Classification Metrics of the Proposed Model

	Precision	Recall	F1-Score	Support
Genuine	0.98	0.99	0.98	8267
Fake	0.97	0.95	0.96	4088
Macro Avg	0.97	0.97	0.97	12355
Weighted Avg	0.97	0.97	0.97	12355

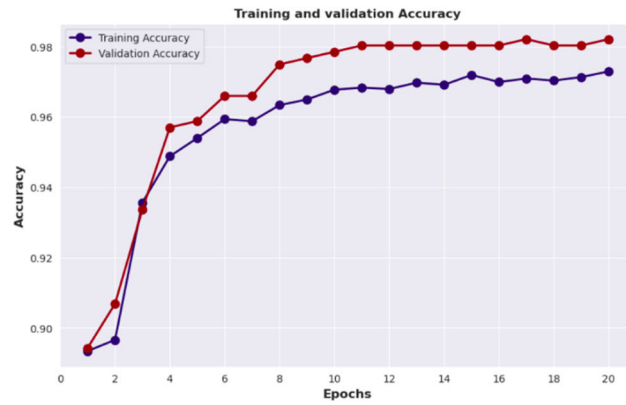

Fig. 7. Training and validation accuracy curve.

Fig. 8. Training and validation loss curve.

A. Experimental Results

Table II depicts the model's performance mentioned above in terms of accuracy, precision, recall, and $F1$ score for real and fake accounts. Figs. 7 and 8 depict the AUC–ROC curves for accuracy and loss, respectively, during training and testing the proposed model on the datasets, as mentioned earlier. Considering the nature of our issue formulation, we display the confusion matrices in Fig. 9 to assess the classifier's performance and better comprehend its strengths and limitations. Finally, we compare our model's performance and efficiency to current models.

Table II demonstrates that the proposed approach relatively easily distinguishes between fake and genuine accounts. Here,

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

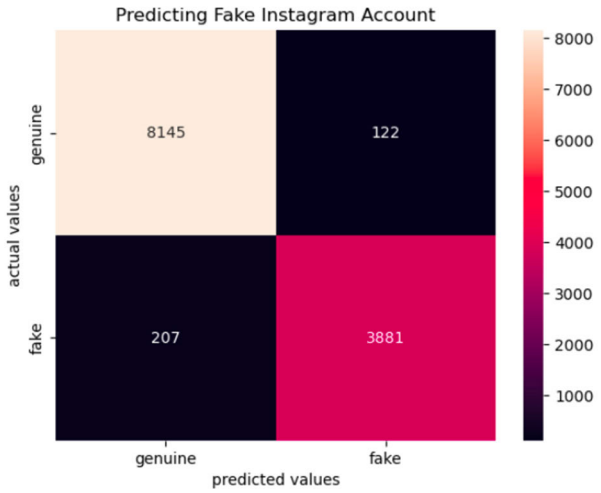

Fig. 9. Confusion matrix.

it indicates that we had supplied 12 355 data samples to the model, 8267 samples related to genuine accounts, whereas 4088 samples related to fake accounts. Despite the imbalanced nature of the dataset, our proposed model classified well with good precision, recall, and $F1$-score. Here, we got 98% precision, 99% recall, and 98% $F1$-score for real accounts, whereas 97% precision, 95% recall, and 96% $F1$-score for fake accounts. The average and overall accuracy of the proposed model is 98%, which is quite impressive.

Fig. 7 depicts the training and validation AUC–ROC curves of the proposed model on the Twitter dataset. We trained and validated this model using different epochs, where the training stabilizes after 20 epochs. We also tested the model with longer epochs up to 100 and varying dropout rates; however, training needed improvement. Nonetheless, the model achieves an accuracy of 99% during training and around 97% during validation.

Fig. 7 shows the training and validating accuracy curves indicating the result accuracy. Here, we achieve a maximum of 99% accuracy during training and 97% accuracy during validation.

Training loss and validation loss are two crucial measures used to evaluate the performance of a machine learning model during the training and validation stages, respectively. They are used to verify that the model adequately matches the training data and is effective with new data. Fig. 8 demonstrates the initial label cost of training and validation lowers until it stabilizes after 20 epochs, after which there is no further improvement. However, the average model compares the actual class labels of cases with those predicted by the model.

Fig. 8 represents the training and validation loss curves indicating the loss of cost of training and validation. Here, we reduce the cost of training and validation from 0.35 to 0.10.

A model's strengths and shortcomings may be determined using the metrics derived from a confusion matrix. A confusion matrix, as seen in Fig. 9, is an effective method for evaluating the performance of a classification cost of training and validation which decreased from 0.35 to 0.10.

TABLE III
COMPARISONS WITH EXISTING MODELS

Methods	Accuracy	Precision	Recall	F1-Score
NB [18]	0.90	0.93	0.88	0.90
LR+PSO [19]	0.96	0.79	0.97	0.89
RNN [20]	0.96	0.95	0.96	0.96
SML [21]	0.95	0.95	0.93	0.94
Proposed	**0.97**	**0.97**	**0.97**	**0.97**

Fig. 9 demonstrates the confusion matrix indicating 12 355 accounts, of which 8267 are genuine accounts and 4088 are fake accounts. It shows that out of 8267 genuine accounts, 8145 are predicted as true positive (TP), and the rest, 122, fall under false negative (FN). In terms of fake accounts, out of 4088 accounts, 207 are false positive (FP) and 3881 are true negative (TN). These results show the correctly predicted genuine accounts, indicating 8145 as correctly classified genuine accounts and 3881 as correctly classified as fake accounts.

B. Comparative Analysis

In computing, comparison and efficiency are key ideas for figuring out how well a model works and how to improve it. With an average accuracy of 90%–96% and a high FP rate on unimodal datasets of other models, our proposed model receives an average accuracy of 97%, which is higher than other. Table III shows comparison of the proposed method with other work to highlight its advantage, especially on difficult task.

Although all the methods presented in Table III are difficult to achieve, RNN is more difficult when compared with Naive Bayes (NB), SML, and logistic regression with particle swarm optimization (LR + PSO). RNN uses sequential patterns to identify fake accounts but needs more training and computing time. Learning meaningful patterns requires a lot of labeled training data. NB implies feature independence; however, account-related behaviors may not be independent. This assumption makes it difficult to capture complicated feature interactions and dependencies. Similarly, success of LR + PSO fake account detection relies on input feature quality and relevance by distinguishing real accounts from fake accounts. This requires domain knowledge and careful analysis of the problem that makes it difficult to achieve.

As indicated in Table III, with an average accuracy of 0.97, precision 0.97, recall 0.97, and $F1$-score 0.97, respectively, the proposed method achieved better results in comparison to all other methods covered in this study. Therefore, the suggested model's findings, as described above, are pretty good and more trustworthy than those of the current SotA techniques and other existing approaches. Fig. 10 shows the comparative analysis results in graphical form.

In summary, the suggested method accurately spots fake accounts on social media sites. Compared with earlier algorithms that used text-based characteristics, incorporating multimodal data and deep learning techniques allows for improved detection of bogus accounts. The results show how integrating visual and network-based data may enhance the efficiency of fraudulent account identification. This strategy may be used

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

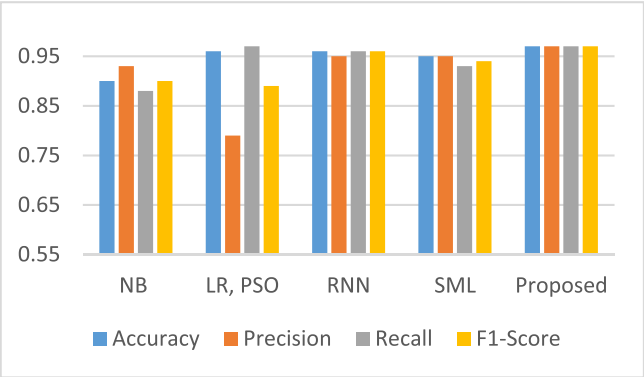

Fig. 10. Graphical comparative analysis.

on many different social media platforms and can enhance the safety and reliability of online communities. The suggested approach is sufficiently adaptable and practical to manage cross-platform hacked social media accounts. Overall, the research makes a significant addition to the study of social media analytics and the identification of bogus accounts.

VI. Conclusion

We proposed a novel method for detecting fake accounts on Twitter using multimodal data with deep learning. Our approach combined textual, visual, and network-based features to capture the various characteristics of fake accounts. It used a deep neural network incorporating CNNs, LSTMs, and graph GCNs to analyze these features. We evaluated our approach on the multimodal Twitter datasets and achieved 99% accuracy during training and around 97% during validation, which is higher than expected. The cost of training and validation decreased from 0.35 to 0.10. In future work, we plan to explore other types of data, such as temporal data and user behavior data, to further improve the accuracy of fake account detection. Overall, our proposed method provides a promising direction for detecting fake accounts on social media using multimodal data with deep learning.

References

[1] U. Can and B. Alatas, "A new direction in social network analysis: Online social network analysis problems and applications," *Phys. A, Stat. Mech. Appl.*, vol. 535, Dec. 2019, Art. no. 122372.

[2] X. Chen, Y. Yuan, L. Lu, and J. Yang, "A multidimensional trust evaluation framework for online social networks based on machine learning," *IEEE Access*, vol. 7, pp. 175499–175513, 2019.

[3] A. Bovet and H. A. Makse, "Influence of fake news in Twitter during the 2016 U.S. presidential election," *Nature Commun.*, vol. 10, no. 1, p. 7, Jan. 2019.

[4] B. Wang, N. Z. Gong, and H. Fu, "GANG: Detecting fraudulent users in online social networks via guilt-by-association on directed graphs," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2017, pp. 465–474.

[5] U. Arora, H. S. Dutta, B. Joshi, A. Chetan, and T. Chakraborty, "Analyzing and detecting collusive users involved in blackmarket retweeting activities," *ACM Trans. Intell. Syst. Technol.*, vol. 11, no. 3, pp. 1–24, Jun. 2020.

[6] W. Zhou, J. Wen, M. Gao, H. Ren, and P. Li, "Abnormal profiles detection based on time series and target item analysis for recommender systems," *Math. Problems Eng.*, vol. 2015, pp. 1–9, May 2015.

[7] W. Zeng, R. Tang, H. Wang, X. Chen, and W. Wang, "User identification based on integrating multiple user information across online social networks," *Secur. Commun. Netw.*, vol. 2021, pp. 1–14, May 2021.

[8] M. Mohammadrezaei, M. E. Shiri, and A. M. Rahmani, "Identifying fake accounts on social networks based on graph analysis and classification algorithms," *Secur. Commun. Netw.*, vol. 2018, pp. 1–8, Aug. 2018.

[9] Z. Yang, X. Chen, H. Wang, W. Wang, Z. Miao, and T. Jiang, "A new joint approach with temporal and profile information for social bot detection," *Secur. Commun. Netw.*, vol. 2022, pp. 1–14, May 2022.

[10] M. Siino, E. Di Nuovo, I. Tinnirello, and M. La Cascia, "Fake news spreaders detection: Sometimes attention is not all you need," *Information*, vol. 13, no. 9, p. 426, Sep. 2022.

[11] F. Bertini, R. Sharma, and D. Montesi, "Are social networks watermarking us or are we (unawarely) watermarking ourself?" *J. Imag.*, vol. 8, no. 5, p. 132, May 2022.

[12] M. Egele, G. Stringhini, C. Kruegel, and G. Vigna, "Towards detecting compromised accounts on social networks," *IEEE Trans. Depend. Secure Comput.*, vol. 14, no. 4, pp. 447–460, Jul. 2017.

[13] Z. Wu, G. Li, Q. Liu, G. Xu, and E. Chen, "Covering the sensitive subjects to protect personal privacy in personalized recommendation," *IEEE Trans. Services Comput.*, vol. 11, no. 3, pp. 493–506, May 2018.

[14] Z. Qu, C. Lyu, and C.-H. Chi, "MUSH: Multi-stimuli Hawkes process based Sybil attacker detector for user-review social networks," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 4, pp. 4600–4614, Dec. 2022.

[15] C. Lin, S. Chen, M. Zeng, S. Zhang, M. Gao, and H. Li, "Shilling black-box recommender systems by learning to generate fake user profiles," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 24, 2022, doi: 10.1109/TNNLS.2022.3183210.

[16] S. H. Moghaddam and M. Abbaspour, "Friendship preference: Scalable and robust category of features for social bot detection," *IEEE Trans. Depend. Secure Comput.*, vol. 20, no. 2, pp. 1516–1528, Mar. 2023.

[17] N. S. G. Bharti and P. Gulia, "Exploring machine learning techniques for fake profile detection in online social networks," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 13, no. 3, pp. 2962–2971, 2023.

[18] B. Ersahin, Ö. Aktas, D. Kilinç, and C. Akyol, "Twitter fake account detection," in *Proc. Int. Conf. Comput. Sci. Eng. (UBMK)*, Oct. 2017, pp. 388–392.

[19] K. K. Bharti and S. Pandey, "Fake account detection in Twitter using logistic regression with particle swarm optimization," *Soft Comput.*, vol. 25, no. 16, pp. 11333–11345, Aug. 2021.

[20] B. S. Borkar, D. R. Patil, A. V. Markad, and M. Sharma, "Real or fake identity deception of social media accounts using recurrent neural network," in *Proc. Int. Conf. 4th Ind. Revolution Based Technol. Practices (ICFIRTP)*, Nov. 2022, pp. 80–84.

[21] O. Prakash and R. Kumar, "Fake account detection in social networks with supervised machine learning," in *Proc. Int. Conf. IoT, Intell. Comput. Secur.* Cham, Switzerland: Springer, 2023, pp. 287–295.

[22] M. K. Hasan et al., "Evolution of industry and blockchain era: Monitoring price hike and corruption using BIoT for smart government and Industry 4.0," *IEEE Trans. Ind. Informat.*, vol. 18, no. 12, pp. 9153–9161, Dec. 2022.

[23] A. Diro, H. Reda, N. Chilamkurti, A. Mahmood, N. Zaman, and Y. Nam, "Lightweight authenticated-encryption scheme for Internet of Things based on publish-subscribe communication," *IEEE Access*, vol. 8, pp. 60539–60551, 2020.

[24] I. Ahmed, M. Ahmad, and G. Jeon, "Social distance monitoring framework using deep learning architecture to control infection transmission of COVID-19 pandemic," *Sustain. Cities Soc.*, vol. 69, Jun. 2021, Art. no. 102777.

[25] I. Ahmed, M. Ahmad, J. J. P. C. Rodrigues, G. Jeon, and S. Din, "A deep learning-based social distance monitoring framework for COVID-19," *Sustain. Cities Soc.*, vol. 65, Feb. 2021, Art. no. 102571.

[26] G. Zhang, J. Wu, G. Jeon, Y. Chen, Y. Wang, and M. Tan, "Towards understanding metaverse engagement via social patterns and reward mechanism: A case study of *Nova empire*," *IEEE Trans. Computat. Social Syst.*, early access, Nov. 24, 2022, doi: 10.1109/TCSS.2022.3211679.

[27] I. Priyadarshini, S. Sahu, and R. Kumar, "A transfer learning aproach for detecting offensive and hate speech on social media platforms," *Multimedia Tools Appl.*, vol. 82, pp. 27473–27499, Feb. 2023.

[28] T. Vo et al., "Crime rate detection using social media of different crime locations and Twitter part-of-speech tagger with Brown clustering," *J. Intell. Fuzzy Syst.*, vol. 38, no. 4, pp. 4287–4299, Apr. 2020.

[29] C. L. Kwek, K. S. Yeow, L. Zhang, K. H. Keoy, and G. Japos, "The determinants of fake news adaptation during COVID-19 pandemic: A social psychology approach," *Recoletos Multidisciplinary Res. J.*, vol. 10, no. 2, pp. 19–39, Dec. 2022.

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

[30] P. Tiwari, A. Lakhan, R. H. Jhaveri, and T.-M. Gronli, "Consumer-centric Internet of Medical Things for cyborg applications based on federated reinforcement learning," *IEEE Trans. Consum. Electron.*, early access, Feb. 7, 2023, doi: 10.1109/TCE.2023.3242375.

[31] A. Ghazvini, S. N. H. S. Abdullah, M. K. Hasan, and D. Z. A. B. Kasim, "Crime spatiotemporal prediction with fused objective function in time delay neural network," *IEEE Access*, vol. 8, pp. 115167–115183, 2020.

[32] K. Ali, C. Li, K. Zain-ul-Abdin, and S. A. Muqtadir, "The effects of emotions, individual attitudes towards vaccination, and social endorsements on perceived fake news credibility and sharing motivations," *Comput. Hum. Behav.*, vol. 134, Sep. 2022, Art. no. 107307.

[33] C. Beauvais, "Fake news: Why do we believe it?" *Joint Bone Spine*, vol. 89, no. 4, Jul. 2022, Art. no. 105371.

[34] J. Wolak, K. J. Mitchell, and D. Finkelhor, "Does online harassment constitute bullying? An exploration of online harassment by known peers and online-only contacts," *J. Adolescent Health*, vol. 41, no. 6, pp. S51–S58, Dec. 2007.

[35] A. Basit, M. Zafar, X. Liu, A. R. Javed, Z. Jalil, and K. Kifayat, "A comprehensive survey of AI-enabled phishing attacks detection techniques," *Telecommun. Syst.*, vol. 76, no. 1, pp. 139–154, Jan. 2021.

[36] D. Saraswat, F. Patel, P. Bhattacharya, A. Verma, S. Tanwar, and R. Sharma, "UpHaaR: Blockchain-based charity donation scheme to handle financial irregularities," *J. Inf. Secur. Appl.*, vol. 68, Aug. 2022, Art. no. 103245.

[37] L. Chen, W. Li, H. Chen, and S. Geng, "Detection of fake reviews: Analysis of sellers' manipulation behavior," *Sustainability*, vol. 11, no. 17, p. 4802, Sep. 2019.

[38] P. N. Petratos, "Misinformation, disinformation, and fake news: Cyber risks to business," *Bus. Horizons*, vol. 64, no. 6, pp. 763–774, Nov. 2021.

[39] M. Simmons and J. S. Lee, "Catfishing: A look into online dating and impersonation," in *Social Computing and Social Media. Design, Ethics, User Behavior, and Social Network Analysis*. Copenhagen, Denmark: Springer, Jul. 2020, pp. 349–358.

[40] Y. Boshmaf et al., "Íntegro: Leveraging victim prediction for robust fake account detection in large scale OSNs," *Comput. Secur.*, vol. 61, pp. 142–168, Aug. 2016.

[41] M. Vyawahare and S. Govilkar, "Fake profile recognition using profanity and gender identification on online social networks," *Social Netw. Anal. Mining*, vol. 12, no. 1, p. 170, Dec. 2022.

[42] S. Adikari and K. Dutta, "Identifying fake profiles in LinkedIn," 2020, *arXiv:2006.01381*.

[43] J. B. Munga and P. Mohandas, "Feature selection for identification of fake profiles on Facebook," in *Proc. 6th Kuala Lumpur Int. Conf. Biomed. Eng. (BioMed)*. Cham, Switzerland: Springer, Jul. 2022, pp. 489–497.

[44] M. Orabi, D. Mouheb, Z. Al Aghbari, and I. Kamel, "Detection of bots in social media: A systematic review," *Inf. Process. Manage.*, vol. 57, no. 4, Jul. 2020, Art. no. 102250.

[45] K. S. Adewole, N. B. Anuar, A. Kamsin, K. D. Varathan, and S. A. Razak, "Malicious accounts: Dark of the social networks," *J. Netw. Comput. Appl.*, vol. 79, pp. 41–67, Feb. 2017.

[46] K. S. Adewole, T. Han, W. Wu, H. Song, and A. K. Sangaiah, "Twitter spam account detection based on clustering and classification methods," *J. Supercomput.*, vol. 76, no. 7, pp. 4802–4837, Jul. 2020.

[47] D. Ramalingam and V. Chinnaiah, "Fake profile detection techniques in large-scale online social networks: A comprehensive review," *Comput. Electr. Eng.*, vol. 65, pp. 165–177, Jan. 2018.

[48] O. Prakash and R. Kumar, "Fake news detection in social networks using attention mechanism," in *Proc. Int. Conf. Cogn. Intell. Comput. (ICCIC)*. Cham, Switzerland: Springer, 2023, pp. 453–462.

[49] Q. Ye et al., "Modeling access environment and behavior sequence for financial identity theft detection in e-commerce services," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2022, pp. 1–8.

[50] Y.-C. Wei, Y.-X. Lai, and M.-E. Wu, "An evaluation of deep learning models for chargeback fraud detection in online games," *Cluster Comput.*, vol. 26, pp. 927–943, Jul. 2022.

[51] J. Roesslein. (2009). *Tweepy Documentation*. [Online]. Available: http://tweepy.readthedocs.io/en/

[52] J. Xie et al., "Advanced dropout: A model-free methodology for Bayesian dropout optimization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 4605–4625, Sep. 2022.

[53] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Mach. Learn. (ICML)*, 2015, pp. 448–456.

Bharti Goyal received the B.Sc. and M.Sc. degrees in computer science from Maharani Kishori Kanya Mahavidyalaya Hodal Palwal (Haryana), Maharshi Dayanand University, Rohtak, Haryana, India, in 2015 and 2017, respectively, where she is currently pursuing the Ph.D. degree with the Department of Computer Science and Applications.

Nasib Singh Gill received the Ph.D. degree in computer science from the University of Delhi, New Delhi, India, in 1996, and the Post-Doctoral Research degree in computer science from the Brunel University, West London, Uxbridge, U.K., during 2001–2002, and he received the MBA (IT) degree from Sikkim Manipal University, Gangtok, in July 2009.

He is a recipient of the Commonwealth Fellowship Award of the British Government for the year 2001. He is currently the Head of the Department of Computer Science and Applications, Maharshi Dayanand University, Rohtak, India. He is also working as the Director, Directorate of Distance Education and Digital Learning Centre, Maharshi Dayanand University, Rohtak. He has published more than 304 research articles and authored five popular books. He has so far guided 12 Ph.D. scholars and guiding about five more scholars. His research interests primarily include IoT, machine and deep learning, information and network security, data mining and data warehousing, natural language processing (NLP), measurement of component-based systems, etc.

Dr. Gill is an active professional member of The Institute of Electronics and Telecommunication Engineers (IETE), International Association of Engineers (IAENG), and Computer Society of India (CSI).

Preeti Gulia received the Ph.D. degree in computer science from Maharshi Dayanand University, Rohtak, Haryana, in 2013.

She is currently working as an Associate Professor at the Department of Computer Science and Applications, Maharshi Dayanand University, Rohtak, India, since 2009. She has published more than 65 research papers and articles in journals and conferences of national/international repute including Association for Computing Machinery (ACM) and Scopus. Her area of research includes data mining, big data, machine learning, deep learning, IoT, and software engineering.

Dr. Gulia is an active professional member of International Association of Engineers (IAENG), Computer Society of India (CSI), and ACM. She is also serving as an editorial board member and active reviewer for the international/national journals.

Om Prakash is currently pursuing the Ph.D. degree with the School of Computer and Systems Sciences, Jawaharlal Nehru University, New Delhi, India.

He is awarded with Junior Research Fellowship at the National Eligibility Test (UGC-NET) India for his research.

Ishaani Priyadarshini received the M.S. degree in cybersecurity and the Ph.D. degree in artificial intelligence from the University of Delaware, Newark, DE, USA, in 2021 and 2018, respectively.

She is working as a Faculty Member with the School of Information, University of California, Berkeley, Berkeley, USA. She has authored and edited several books for reputed publishers and is also an author of several publications for SCIE-indexed journals. As a certified reviewer, she conducts peer reviews of research articles for prestigious IEEE, Elsevier, and Springer journals and is a part of the editorial board for many international journals. Her areas of research include cybersecurity, artificial intelligence, and machine learning.

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Rohit Sharma (Senior Member, IEEE) is an Associate Professor with the Department of Electronics and Communication Engineering, SRM Institute of Science and Technology, Delhi NCR Campus, Ghaziabad, India. He has received the Young Researcher Award at the "2nd Global Outreach Research and Education Summit & Awards 2019" hosted by the Global Outreach Research & Education Association (GOREA). He is serving as a Guest Editor for the SCI journal of Elsevier, Centre for Environment Education (CEE). He has actively been an organizing end of various reputed international conferences.

Dr. Sharma is an active member of Indian Society for Technical Education (ISTE), IEEE, ICS, International Association of Engineers (IAENG), and International Association of Computer Science and Information Technology (IACSIT). He is an Editorial Board Member and reviewer of more than 12 international journals and conferences, including the topmost journal IEEE ACCESS and IEEE INTERNET OF THINGS JOURNAL. He serves as a Book Editor for seven different titles to be published by CRC Press, Taylor & Francis Group, USA, Apple Academic Press, Springer, etc.

Ahmed J. Obaid received the B.Sc. degree in information systems from the Faculty of Computer Science, University of Anbar, Ramadi, Iraq, in 2005, the M.Tech. degree in computer science and engineering from the Siddaganga Institute of Technology (SIT), Jawaharlal Nehru Technological University Hyderabad, Telangana, India, in 2012, and the Ph.D. degree in web mining and data mining from the University of Babylon, Hillah, Iraq, in 2017.

He is a Full Assistant Professor with the Department of Computer Science, Faculty of Computer Science and Mathematics, University of Kufa, Kufa, Iraq. His main line of research is web mining techniques and applications, image processing in web platforms, genetic algorithms, and information theory. He is an Associate Editor with the *Brazilian Journal of Operations & Production Management* (ESCI), a Guest Editor of the *Key Engineering Material* (KEM, Scopus journal), a Guest Editor of the MAICT-19 and ICMAICT-20 issues in the *IOP Journal of Physics* (Scopus indexed proceeding), a Guest Editor of the *Journal of Engineering Science and Technology* (Scopus, WOS), a Managing Editor of the *American Journal of Business and Operations Research* (AJBOR), and an Associate Editor of the *International Journal of Applied Science and Technology* (Scopus journal). He is also a reviewer in many Scopus journals (Scientific Publication, Taylor and Francis, ESTA, and many others). He is the Leader of ICPAS, MAICT-19, MAICT-20, IHICPS, and IICPS Conferences. He has supervised several final projects for bachelors's and master's in his main line of work. He has edited some books, such as *Advance Material Science and Engineering* (ISBN: 9783035736779, Scientific.net publisher), *Computational Intelligence Techniques for Combating COVID-19* (ISBN: 978-3-030-68936-0 EAI/Springer), and *A Fusion of Artificial Intelligence and Internet of Things for Emerging Cyber Systems* (ISBN: 978-3-030-76653-5 Springer). He has authored and coauthored several scientific publications in journals and conferences and is a frequent reviewer of international journals and international conferences.

Kusum Yadav is currently working with the Department of Information and Computer Science, College of Computer Science and Engineering, University of Haíl, Kingdom of Saudi Arabia. She has more than 15 years of experience in academic and research in addition to three years of industry experience. She has published several papers in SCI/Scopus/high-impact factored journals, international conferences, and book chapters. She has 15 published patents and nine granted copyright. Her research interests include big data, cyber security, machine learning, soft computing, data mining, hybrid systems, and feature selection. She has guided and is guiding M.Sc. (artificial intelligence) students and Ph.D. scholars. She has organized many events such as faculty development programs, hackathons, and workshops.

Dr. Yadav has served as session chair, member of the advisory/technical program committee, and reviewer for many international/national conferences.

Authorized licensed use limited to: Birla Inst of Technology and Science Pilani Dubai. Downloaded on February 18,2026 at 14:34:06 UTC from IEEE Xplore. Restrictions apply.